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PHARMACOKINETICS AND DISPOSITION

Pharmacokinetics of morphine-6-glucuronide following oral administration in healthy volunteers

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Abstract
Aim After oral administration, morphine-6-glucuronide (M6G) displays an atypical absorption profile with two peak plasma concentrations. A proposed explanation is that M6G is hydrolysed to morphine in the colon, which is then absorbed and subsequently undergoes metabolism in the liver to morphine-3-glucuronide (M3G) and M6G. The aims of this study were to confirm and elucidate the biphasic absorption profile as well as clarify the conversion of M6G to morphine after a single oral administration of M6G in healthy volunteers.
Methods The study was conducted accordingly to a non-blinded, randomised, balanced three-way crossover design in eight healthy male subjects. The subjects received 200 mg oral M6G, 50 mg oral M6G and 30 mg oral morphine. Blood samples were collected until 72 h after M6G administration and until 9 h after morphine administration. Paracetamol and sulfasalazine were coadministered with M6G as markers for the gut contents reaching the duodenum and colon, respectively.
Results The plasma concentration peaks of M6G were seen at 4.0 (2.0–6.0) and 18 (12.0–24.0) h after 200 mg M6G and at 3.5 (2.0–6.0) and 21.3 (10.0–23.3) h after 50 mg M6G, which was in agreement with previously published results. The K_{M6G_abs}/K_{M6G_M6G} ratio was found to be 10.
Conclusion The pharmacokinetic profile of M6G after oral administration was confirmed and with the presence of M3G and morphine in plasma after oral administration of M6G, proof seems to be found of the constant and prolonged absorption of M6G. The K_{M6G_abs}/K_{M6G_M6G} ratio of 10 indicates that the second absorption peak of M6G consists of approximately 10 times more absorbed M6G than reglucuronidated M6G. However, further studies are required to determine the precise kinetics of the second absorption peak.

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Introduction

Morphine is considered the gold standard for the treatment of moderate to severe pain despite its many side effects. The active metabolite, morphine-6-glucuronide (M6G), has an analgesic effect equivalent to morphine, but the side effects seem less frequent, particularly nausea, vomiting [1–4] and respiratory depression [2, 5, 6]. After oral administration, M6G displays an atypical absorption profile. Two peak plasma concentrations are observed at different times. The first peak is seen approximately 3–5 h after administration of an oral solution, whereas the second appears after 13–20 h [7, 8]. The late peak is associated with the appearance of both morphine and morphine-3-glucuronide (M3G) in plasma. A possible explanation is that M6G is hydrolysed to morphine in the colon, which is then absorbed and subsequently metabolised in the liver to M3G and M6G [7, 9]. In a study by Chapman, a second peak was seen after approximately 10 h; however, neither morphine nor M3G was detected in plasma [10]. After parenteral administration in humans, it is generally considered that M6G is not metabolised [11–13]. The benefits of M6G in the clinical setting are limited due to a poor bioavailability (9–13%) [7, 8, 10]. However, the side effects seem less frequent, particularly nausea and vomiting [1–6]. The absorption enhancer TWEEN 20 seemed promising with respect to increasing the absorption fraction in in vitro studies [14]. However, coadministration of 500 mg TWEEN 20 did not improve the bioavailability of orally administered M6G [8].

The aims of this study was to confirm the biphasic absorption profile and elucidate the hydrolysis of M6G to morphine after a single oral dose of M6G in healthy volunteers.

Materials and methods

Study design

The study was conducted according to a nonblinded, randomised, balanced three-way crossover design with a 7-day washout period between each M6G administration. The Regional Committee on Biomedical Research Ethics for Frederiksberg and Copenhagen and the Danish Medicines Agency approved the protocol, informed consent form and amendments for the study. Verbal and written information about the study was given to the volunteers, and written informed consent was obtained according to the ethical principles stated in the Declaration of Helsinki II.

Subjects

Study subjects were eight healthy male volunteers aged 18–35 years, all nonsmokers of normal body weight (body mass index between 20 and 25). Exclusion criteria included mental illness; alcohol or drug abuse or addiction; a history of alcohol abuse; illicit drug use or suspected abuse; a positive urine drug screen for methamphetamine, opiates, cocaine metabolites, cannabinoids, phencyclidine, benzodiazepines, barbiturates, methadone, tricyclic antidepressants, and amphetamines; as well as long-term treatment with analgesics or daily use of benzodiazepines. Further, candidates who had received antibiotics within 4 weeks prior to the study; donated blood within 3 months prior to the study; or with deviant echocardiogram (ECG), liver or kidney function [lactic dehydrogenase (LDH), alanine aminotransferase (ALAT), aspartate aminotransferase (ASAT), P-creatinine], coagulation factors II, VII and X, or blood values (haemoglobin, potassium, sodium, alkaline phosphatase) were excluded from entry into the study. Known hypersensitivity to morphine, paracetamol, sulfasalazine, sulphonamide, or salicylate was also an exclusion criterion.

Drugs

M6G 20 mg/ml sterile solution for injection in ampoules of 2 ml was imported from CeNeS Limited, Cambridge, UK and supplied by the hospital pharmacy at Herlev University Hospital, Denmark. Certificate of analysis showed that the batch contained 0.06% morphine and no detectable M3G, i.e. <0.03%. These amounts are not suspected to interfere with subsequent study results. Paracetamol tablets 500 mg, sulfasalazine tablets 500 mg, and morphine tablets 30 mg were also supplied by the hospital pharmacy. Paracetamol was coadministered as a marker for gastric emptying time, whereas sulfapyridine (the hydrolysate of sulfasalazine) was used as a marker for arrival of the gut content to the colon [15]. Each study subject received on different days:

– Preparation 1: M6G 200 mg as an oral solution
– Preparation 2: M6G 50 mg as an oral solution
– Preparation 3: Morphine 30 mg as a tablet.

As the aim of the study was to confirm the biphasic absorption profile, it was decided to repeat the doses used previously [8].

Procedures

Opioids, including codeine and food containing poppy seeds, were prohibited for 72 h and paracetamol for 24 h prior to each protocol day.

Food and fluid restrictions

Fasting from midnight was required prior to all 3 protocol days. During the first 6 h, the study subjects received 500 ml of dextrose 5% as an infusion to prevent hypoglycaemia. Three hours after drug administration, the study subjects were allowed to drink 50 ml of water every second hour until lunch was served 6 h after drug administration. The study subjects were instructed to reproduce the fluid intake precisely on the 2 protocol days when they received the oral formulations of M6G.

Drug administration and blood sampling

M6G was administered as an oral solution, prepared from a sterile solution for injection diluted with 100 ml of Schweppes Bitter Lemon. Paracetamol 1,000 mg and sulfasalazine 500 mg were coadministered as tablets with the oral test solutions. Morphine was administered as a tablet. A peripheral vein contralateral to the glucose administration was used for blood sampling. For M6G pharmacokinetics, blood samples were obtained prior to drug administration, and at 15, 30 and 60 min and 2, 3, 4, 5, 6, 8, 10, 11, 12, 16, 20, 24, 36, 48, 60 and 72 h after administration. For morphine pharmacokinetics, blood samples were collected before and 5, 15, 30, 45, 60 and 90 min and 2, 4, 6, 8 and 10 h after drug administration. The actual sampling times were noted in the individual case report form and subsequently used in the pharmacokinetic analysis. Routine vital signs—ECG including pulse rate, respiratory rate, blood pressure, and oxygen saturation—were obtained just before drug administration and every 15 min until 12 h after drug administration. These parameters were recorded in the individual case report form. Blood samples were drawn into heparinised glass tubes, which were placed on ice until centrifugation at 3,000 rpm for 10 min. Plasma was separated into tubes, which were stored below −20°C until analysis.

Analytic analyses

Quantification of plasma M6G, morphine, M3G, paracetamol and sulfapyridine was done by high-performance liquid chromatography (HPLC) using mass spectrometry (MS) as the detection principle. After addition of the internal standard, hydromorphone, plasma samples and calibration standards in plasma were subject to solid-phase extraction using 1.0 ml of plasma on Oasis HLB SPE cartridges (30 mg). The final eluate was evaporated to dryness under nitrogen at ambient temperature, and the residue was dissolved in 500 µl of eluent A; 25 µl was injected onto the HPLC column. The LC-MS system consisted of an HP 1100 series chromatograph (Hewlett-Packard, Palo Alto, CA, USA) equipped with a Quatpump, a degasser, column oven (colcomp), an autosampler (ALS), a DAD UV detector operated at 280 nm (bandwidth 16), and a mass selective detector (MSD). Data were collected using the HP ChemStaion software, version 6.03 (Hewlett-Packard). The analytical column was a reversed-phase Zorbax Eclipse XDB (Rockland Technologies, Newport, DE, USA) C8 column (4.6 mm I.D. × 150 mm, 3.5 µm particles). A linear gradient system was applied. Eluent A consisted of 1% formic acid in MeCN:water (3:97 v/v) and eluent B 1% formic acid in MeCN:water (27:73 v/v). The linear gradient was applied from 100% A to 100% B over 4 min; 100% B was maintained for 3 min and then returned to 100% A over 0.5 min. The flow rate was 0.5 ml/min. The MSD detector was equipped with an electrospray interface and used in positive mode for selected ion monitoring (SIM) detection of the four masses: 286.1 (morphine and the internal standard hydromorphone, which are well separated in the HPLC system), 462.2 (M6G and M3G, which are well separated in the HPLC system), 152.1 (paracetamol) and 250.1 (sulfapyridine). The fragmentor voltage was set to 100 V for the mass 286.1 and 120 V for the masses 152.1, 250.1 and 462.2. The voltage over the capillary was kept at 3,000 V. The temperature of the nitrogen drying gas was set to 350°C, with a flow rate of 13 l/min. The nebulizer gas pressure was 30 psi. Limit of quantification (LOQ) for morphine and M3G/M6G was 0.5 nmol/l and 1.0 nmol/l, respectively. Linearity of the calibration curve was proven from 0.5 to 50.000 nmol/l. This was used to detect M6G. Narrow ranges were used to detect morphine and M3G 0.5–50 nmol/l and 0.5–1,000 nmol/l, respectively.

Data analysis

Pharmacokinetic analysis

Individual plasma concentrations for M6G were analysed using WinNonLin, Version 3.3 (Pharsight Corporation, Mountain View, CA, USA). The following pharmacokinetic parameter estimates were determined using noncompartmental analysis. From the data obtained after oral administration, area under the curve $(AUC)_{0-24}$, AUC_{0-72} and lag time [the time prior to the time corresponding to the first measurable (nonzero) concentration (t_{lag})] were determined. Maximum plasma concentration (C_{max}) and time to reach maximum plasma concentration (t_{max}) were registered as the actual obtained values by visual inspection of the individual curves. As M6G is absorbed biphasically, the first t_{max} was to correspond to the first C_{max}. The second t_{max} was to correspond to the C_{max} observed after sulfapyridine was detected in plasma, e.g. when M6G was expected to have reached the colon.

The second absorption peak seen after oral administration of M6G is believed to consist of M6G directly absorbed after oral administration and M6G produced by glucuronidation of morphine. In order to determine the ratio of reglucuronidated M6G to absorbed M6G, the following hypotheses are made:

1. The formation of M3G after M6G administration (M3G_M6G) has to equal the one after morphine administration (M3G_MOR), as M6G is not able to interconvert to M3G, which can then only can be produced by glucuronidation of morphine.

2. Likewise, the M6G formed by glucuronidation of morphine (M6G_M6G) must equal the M6G formed after morphine administration (M6G_MOR).
3. The formation of M6G, which makes up the second absorption peak (M6G_app), consists of directly absorbed M6G (M6G_abs) as well as formation of M6G by glucuronidation of morphine (M6G_M6G).

According to hypothesis 1 and 2, the subsequent equation follows:

$$K_{M3G\_MOR}/K_{M6G\_MOR} = K_{M3G\_M6G}/K_{M6G\_M6G} \quad (1)$$

The different rate constants of M3G and M6G after morphine administration (K_{M3G_MOR} /K_{M6G_MOR}) and M3G after M6G administration (K_{M3G_M6G}) are determined by using the method of residuals [16]. K_{M6G_M6G} can be determined by rearranging Eq. 1.

The M6G rate constant (K_{M6G_app}) hypothesised in hypothesis 3 can also be determined by using the method of residuals on the second absorption peak. K_{M6G_M6G} are given by Eq. 1 and K_{M6G_abs} can be determined by rearranging Eq. 2:

$$K_{M6G\_M6G} + K_{M6G\_abs} = K_{M6G\_app} \quad (2)$$

Statistical analysis

GraphPad Prism version 4.00 for Windows (GraphPad Software, San Diego, CA, USA) was used for all statistical analyses. A paired *t* test with two-tail *P* value was used to test differences in normally distributed data (dose-normalised C_{max} and AUC) between the two oral treatment means. Normality was tested by test for Gaussian distribution. Wilcoxon matched pairs signed rank sum test with two-tail *P* value was used to test differences in nonnormally distributed data (t_{max}). A *P* value of 0.05 was considered statistically significant.

Results

All results are presented as mean values with standard deviation (SD) or 95% confidence interval (95% CI) or as median with range. Mean age, height and weight of the study subjects was 25±5 years, 182±2.7 cm, and 75.4±5.5 kg (BMI 23±1.8), respectively. All study subjects completed all three arms of the study. Side effects were mild to moderate and did not require medical attention.

Pharmacokinetic analysis

Oral administration of M6G

Figure 1 shows individual plasma concentrations versus time profiles of M6G after oral administration. There was considerable intervariability between subjects when M6G was administered orally (see Table 1).

Two plasma concentration peaks were seen at 4.0 (2.0–6.0) and 18 (12.0–24.0) h after 200 mg M6G and at 3.5 (2.0–6.0) and 21.3 (10.0–23.3) h after 50 mg M6G; however, neither of the time differences was statistically significant (*P*=0.3 and 0.8, respectively). The C_{max} corresponding to the first and second t_{max} after 200 mg M6G was 323±194 and 214±75 nmol/l, respectively, and 76±58 and 53±16 nmol/l, respectively, after 50 mg M6G. The $AUC_{0–72}$ after 200 mg M6G and after 50 mg M6G was 7,639±1,762 nmol/l h and 1,903±473 nmol/l h, respectively. When the AUCs were dose-normalised, no statistically significant difference was seen (*P*=0.85).

Paracetamol, which was coadministered as a marker for arrival time in the small intestine, was detectable in plasma

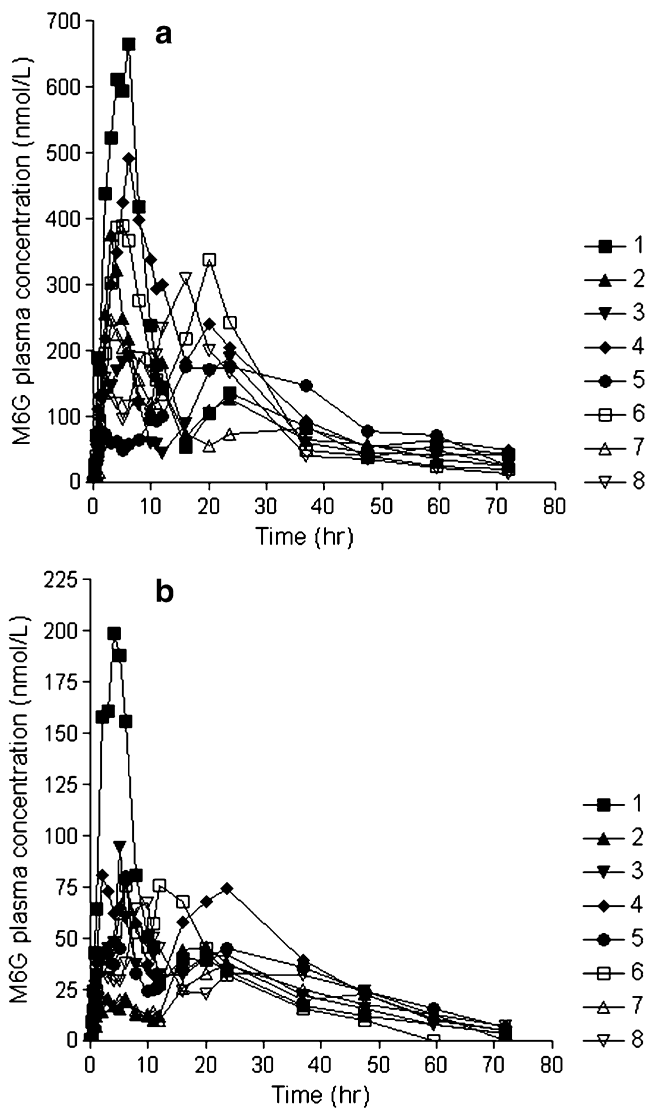

Fig. 1 Individual morphine-6-glucuronide (M6G) concentration after oral administration of M6G 200 mg (**a**) and 50 mg (**b**)

Table 1 Pharmacokinetic parameters of morphine-6-glucuronide (M6G)

Dose (mg)	Oral 50	Oral 200	Significance
T_{max1} (h)	3.5 (2.0–6.0)	4.0 (2.0–6.0)	No (P=0.3)
T_{max2} (h)	21.3 (10.0–23.3)	18 (12.0–24.0)	No (P=0.8)
C_{max1} (nmol/l)	76±58	323±194	
C_{max2} (nmol/l)	53±16	214±75	
$AUC_{0–24}$ (nmol/l/h)	983±398	4317±1602	
$AUC_{0–24}$/dose (nmol/l/h/mg)	20±8	22±8	
$AUC_{0–72}$ (nmol/l/h)	1903±473	7639±1762	
$AUC_{0–72}$/dose (nmol/l/h/mg)	38±10	38±9	No (P=0.85)

T_{max} time to reach maximum plasma concentration, C_{max} maximum plasma concentration, *AUC* area under the curve
Parametric data are presented as mean ± standard deviation (SD), whereas nonparametric data are presented as median (range)

at the first sampling time (15 min) in all study subjects after administration of both 50 mg and 200 mg M6G. T_{max} was 2.0±1.1 and 1.4±0.8 h after coadministration with 50 mg and 200 mg M6G, respectively, which is in accordance with the first t_{max} seen after M6G administration. Arrival time in the colon was to be determined by the time of the first appearance of sulfapyridine in plasma. The t_{lag} for sulfapyridine, which indicates the time for the gut content to reach the colon, varied considerably, from 15 min to 5 h. M3G and morphine was found to appear in plasma concurrently with sulfapyridine. Consequently, high variance in t_{lag} for morphine was seen (16 min–12 h). T_{max} for sulfapyridine was 20±4 and 22±3 h after coadministration with 50 mg and 200 mg M6G, respectively, which is in accordance with the second t_{max} seen after M6G administration. The mean plasma concentration profiles of M6G, M3G, morphine, paracetamol and sulfapyridine are shown in Fig. 2.

Oral administration of morphine

T_{lag} for orally administered morphine was 6±11 min. Five of the eight subjects had detectable concentrations of morphine in the first sample after administration. Time to C_{max} was 56±40 min, and the AUC was 119 (73–181) nmol/l h. Four of eight study subjects (nos. 4, 5, 6 and 7) had a washout period of only 5 days between treatment with M6G 200 mg and treatment with morphine 30 mg. Of these four, three had relatively large concentrations of M6G, M3G and morphine (only two) in the sample taken before morphine administration and the first sample taken after (corresponding to 96 h on Fig. 3). The contribution of the residual area, however, accounts for only 2–5% of the AUC of M6G formed after morphine administration calculated by extrapolating the curve for M6G. Similarly for morphine, the contribution of morphine from M6G 200 mg only accounts for 3% of the total AUC. The contribution from M6G 200 mg is considered negligible and is not accounted for in the subsequent calculations.

The $AUCs_{0–9}$ of M6G and M3G formed after morphine were 815 (730–1,147) and 6,009 (4,552–7,232) nmol/l h, respectively, giving an M3G/M6G ratio of 7.4. The rate constant of M3G and M6G after morphine administration (K_{M3G_MOR} and K_{M6G_MOR}) was determined to be 0.038± 0.0168 min^{-1} and 0.0355±0.0155 min^{-1}, respectively. K_{M3G_M6G} and K_{M6G_app} were 0.0014±0.0006 min^{-1} and 0.0160±0.0176 min^{-1}, respectively. The K_{M6G_abs} was determined to be 0.0145, giving a K_{M6G_abs}/K_{M6G_M6G} ratio of approximately 10.

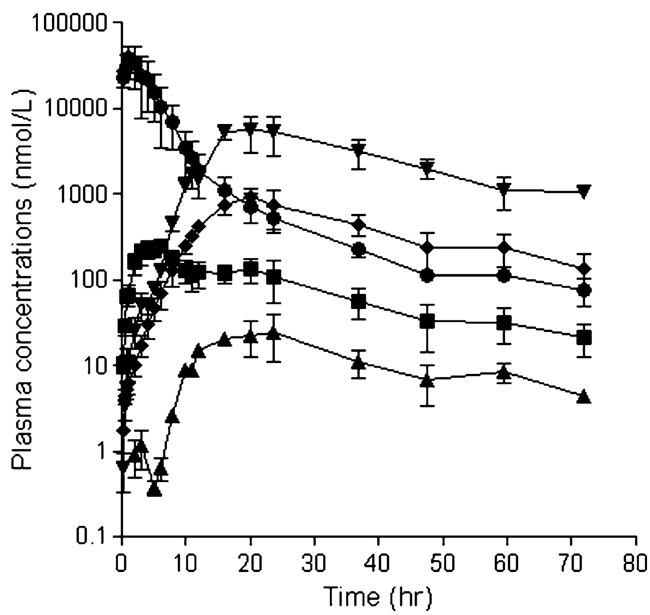

Fig. 2 Mean plasma concentration profiles of morphine-6-glucuronide (M6G) (*square*), morphine-3-glucuronide (M3G) (*diamond*), morphine (*triangle*), paracetamol (*circle*) and sulfapyridine (*invert triangle*) after oral administration of 200 mg M6G

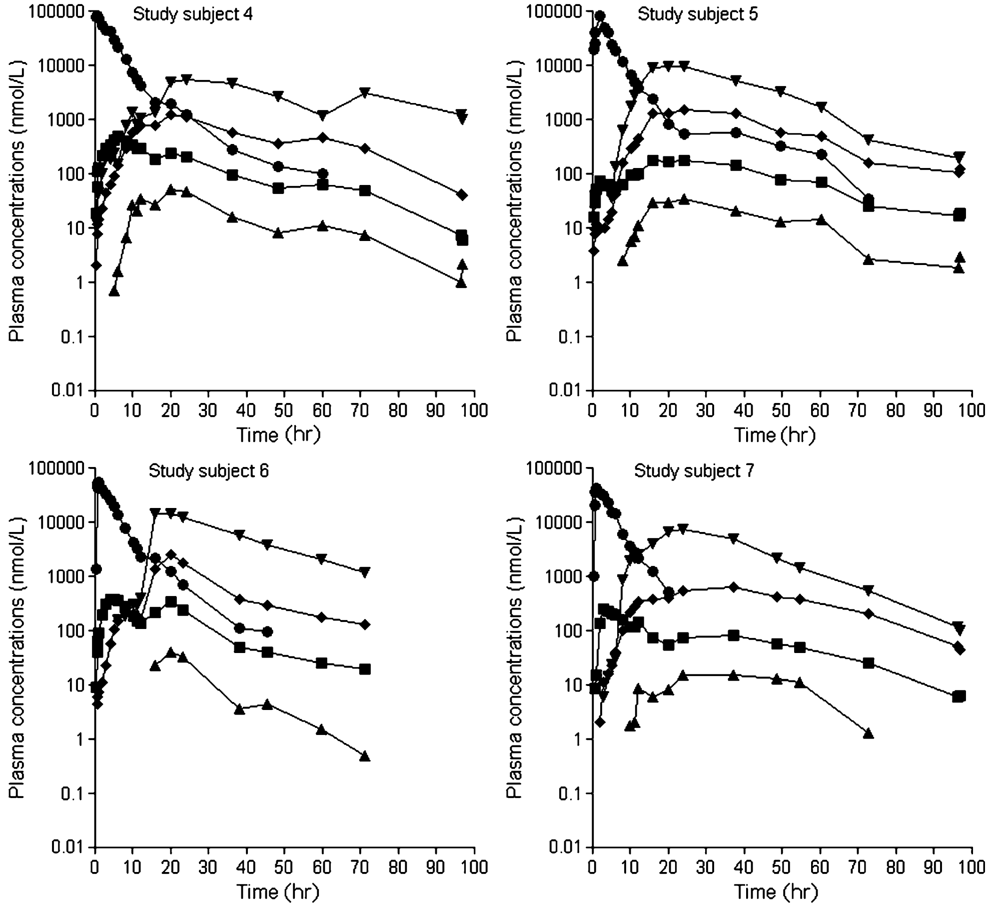

Fig. 3 Individual plasma concentration profiles of morphine-6-glucuronide (M6G) (*square*), morphine-3-glucuronide (M3G) (*diamond*), morphine (*triangle*), paracetamol (*circle*) and sulfapyridine (*invert triangle*) after oral administration of 200 mg M6G for four of the study subjects

Discussion

The pharmacokinetic parameters estimated after oral administration were also within the range of previously published data [7, 8]. Two separate plasma concentration peaks were seen after oral administration. The second plasma concentration peak has been explained by hydrolysis of M6G to morphine, which then undergoes glucuronidation to M6G and M3G, hence the second peak. The occurrence of both M3G and morphine concurrently with the presence of sulfapyridine in plasma supports the theory that M3G is not able to interconvert to M6G and hence has to be produced by glucuronidation of morphine in the liver. In the present study, the plasma concentration peaks after 50 mg occurred slightly later than previously published by Villesen et al. (t_1: 3.5 vs. 2.5; t_2: 21.3 vs. 18 h, respectively). However, the differences are not statistically significant (t_1: P=0.4; t_2: P=0.5). For both C_{max}s and AUCs, linearity is seen between the two doses.

The second plasma concentration peak after 200 mg M6G was seen 18 h after administration, with a corresponding C_{max} of 214±75 nmol/l. In four study subjects, the washout period was only 5 days, and more than 96 h after drug administration, M6G, M3G and morphine were still detectable in plasma in three of four study subjects. All three had relatively high plasma concentrations of sulfapyridine (Fig. 3) at 96 h.

The K_{M6G_abs}/K_{M6G_M6G} ratio of 10 indicates that the second absorption peak of M6G consists of approximately 10× more absorbed M6G than reglucuronidated M6G. This means that the absorption of M6G happens throughout the entire gastrointestinal tract. The clinical consequense is that orally administered M6G acts as a physiological sustained-release system in itself, as earlier purposed by Stain-Texier et al. in 1998 [9]. However, great variability exists, and further pharmacokinetic modelling will be required.

Conclusion

The pharmacokinetic profile of M6G after oral administration was confirmed, and with the presence of M3G and morphine in plasma after administration of M6G seems to be proof of the constant and prolonged absorption of M6G. The K_{M6G_abs}/K_{M6G_M6G} ratio of 10 indicates that the second absorption peak of M6G consists of approximately 10 times more absorbed M6G than reglucuronidated M6G. However, further studies are required to determine the precise kinetics of the second absorption peak.

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